

## 2.8 Product Rule: worked solutions

1.  $f(x) = (2x - 3) \sin x$

$$(uv)' = u'v + uv'$$

Let:

- $u = 2x - 3 \quad \rightarrow \quad u' = 2$
- $v = \sin x \quad \rightarrow \quad v' = \cos x$

So:

$$f'(x) = u'v + uv'$$

$$f'(x) = (2)(\sin x) + (2x - 3)(\cos x)$$

$$f'(x) = 2 \sin x + (2x - 3) \cos x$$

2.  $g(x) = 2x^3 e^x$

$$(uv)' = u'v + uv'$$

Let:

- $u = 2x^3 \quad \rightarrow \quad u' = 6x^2$
- $v = e^x \quad \rightarrow \quad v' = e^x$
- $u' = 2 \cdot 3x^2 = 6x^2$  (power rule)

So:

$$g'(x) = u'v + uv'$$

$$g'(x) = (6x^2)(e^x) + (2x^3)(e^x)$$

$$g'(x) = 6x^2 e^x + 2x^3 e^x$$

$$g'(x) = 2x^2 e^x (3 + x)$$

$$g'(x) = 6x^2 e^x + 2x^3 e^x = 2x^2 e^x (x + 3)$$

3.  $h(x) = 4\sqrt{x} \ln x = 4x^{1/2} \ln x$

$$(uv)' = u'v + uv'$$

Let:

- $u = 4x^{1/2} \quad \rightarrow \quad u' = 2x^{-1/2}$
- $v = \ln x \quad \rightarrow \quad v' = \frac{1}{x}$
- $u' = 4 \cdot \frac{1}{2}x^{1/2-1} = 2x^{-1/2}$

So:

$$h'(x) = u'v + uv'$$

$$h'(x) = \left(2x^{-1/2}\right)(\ln x) + \left(4x^{1/2}\right)\left(\frac{1}{x}\right)$$

$$h'(x) = \frac{2 \ln x}{\sqrt{x}} + 4x^{1/2-1}$$

$$h'(x) = \frac{2 \ln x}{\sqrt{x}} + \frac{4}{\sqrt{x}}$$

$$h'(x) = \frac{2 \ln x + 4}{\sqrt{x}}$$

8.  $s(t) = t^2 \sin t$

$$(uv)' = u'v + uv'$$

Let:

- $u = t^2 \quad \rightarrow \quad u' = 2t$
- $v = \sin t \quad \rightarrow \quad v' = \cos t$

So:

$$s'(t) = u'v + uv'$$

$$s'(t) = (2t)(\sin t) + (t^2)(\cos t)$$

$$s'(t) = 2t \sin t + t^2 \cos t$$

9.  $d(t) = 3\sqrt{t} \ln t = 3t^{1/2} \ln t$

$$(uv)' = u'v + uv'$$

Let:

- $u = 3t^{1/2} \rightarrow u' = \frac{3}{2}t^{-1/2}$
- $v = \ln t \rightarrow v' = \frac{1}{t}$
- $u' = 3 \cdot \frac{1}{2}t^{1/2-1} = \frac{3}{2}t^{-1/2}$

So:

$$d'(t) = u'v + uv'$$

$$d'(t) = \left(\frac{3}{2}t^{-1/2}\right)(\ln t) + \left(3t^{1/2}\right)\left(\frac{1}{t}\right)$$

$$d'(t) = \frac{3 \ln t}{2\sqrt{t}} + 3t^{1/2-1}$$

$$d'(t) = \frac{3 \ln t}{2\sqrt{t}} + \frac{3}{\sqrt{t}}$$

$$d'(t) = \frac{3 \ln t}{2\sqrt{t}} + \frac{6}{2\sqrt{t}}$$

$$d'(t) = \frac{3 \ln t + 6}{2\sqrt{t}} = \frac{3(\ln t + 2)}{2\sqrt{t}}$$